Chapter 17 · Semester 2

MAGNETIC FIELDS

Notes by Cikgu Suhaili

19 worked examples 5 simulations Exercises with answers

17.1 Magnetic field

Magnetic field

a **region of space where a magnetic force is experienced** by a magnetic material, a current-carrying conductor or a moving charge. It is a field of force produced by **permanent magnets or by current-carrying conductors**.

- A magnet has two poles, north (N) and south (S). **Like poles repel, unlike poles attract.**
- The direction of the field at a point is the direction in which the **north pole of a compass needle** points when placed there.
- Field patterns are shown with iron filings or a plotting compass.

Magnetic field lines

- Field lines go **from N to S outside** a magnet (and S to N inside it), so they form closed loops.
- The tangent to a line at a point gives the direction of the field there.
- Field lines **never cross** (the field has only one direction at each point).
- The closer the lines, the stronger the field: the number of lines per unit area is proportional to the field strength.



Figure 17.1: a uniform field and a non-uniform field.



Figure 17.2: a field perpendicular to the page.

Field patterns

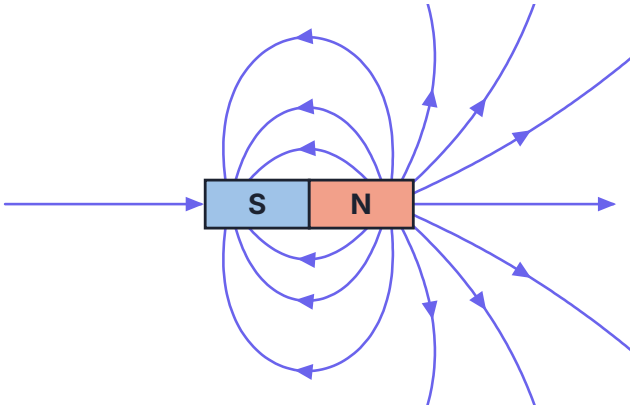


Figure 17.3: a bar magnet.

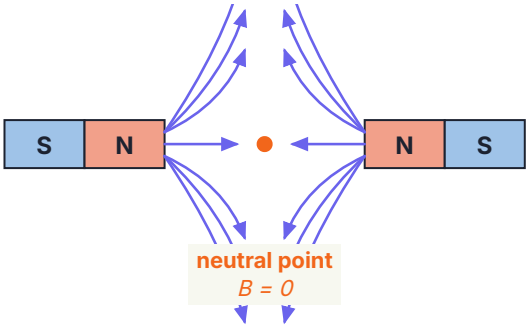


Figure 17.4: two like poles facing each other. At the neutral point the two fields cancel.

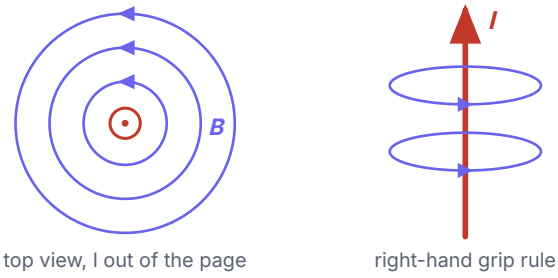


Figure 17.5: a long straight wire: concentric circles. Right-hand grip rule.

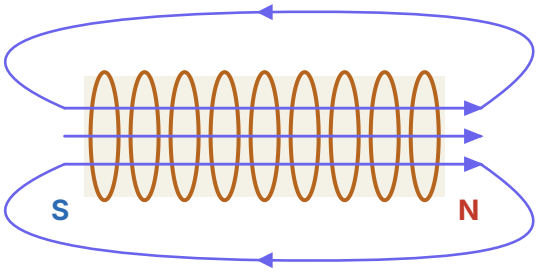


Figure 17.6: a solenoid: like a bar magnet outside, uniform inside.

- **Long straight wire**: concentric circles centred on the wire, further apart away from the wire (weaker field). Direction: **right-hand grip rule**, thumb along the current, fingers curl in the direction of B.
- **Flat circular coil**: near the centre the lines are almost straight and perpendicular to the coil; the coil behaves like a short magnet.
- **Solenoid**: outside, like a bar magnet; inside, the field is **uniform** and parallel to the axis. The end where the current flows anticlockwise (seen from outside) is a north pole.
- The **Earth** has a field like a large bar magnet whose south pole is near the geographic north pole, so a compass needle points north.

Teacher's note

Magnetic flux Φ = **B**·**A** is in Chapter 18 (electromagnetic induction), where the STPM syllabus puts it.

17.2 Force on a moving charge

A charge q moving with velocity **v** in a magnetic field **B** feels a force that is **perpendicular to both v and B**.

$$\mathbf{F} = q\mathbf{v} \times \mathbf{B} \qquad F = qvB\sin\theta$$

θ = angle between v and B

Magnetic flux density, B

defined by $F = qvB\sin\theta$: $B = \dfrac{F}{qv\sin\theta}$. One tesla is the flux density in which **a charge of 1 C moving at 1 m s^{-1} perpendicular to the field feels a force of 1 N**.

- Unit: tesla (T) = N A^{-1} m^{-1} = Wb m^{-2}. B is a vector, along the field lines.
- $F = 0$ when the charge is at rest ($v = 0$) or moves parallel to B (θ = 0° or 180°).
- F is greatest, qvB, when the charge moves perpendicular to B.

Direction: Fleming's left-hand rule

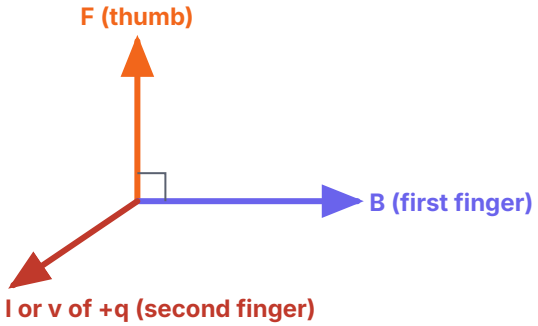


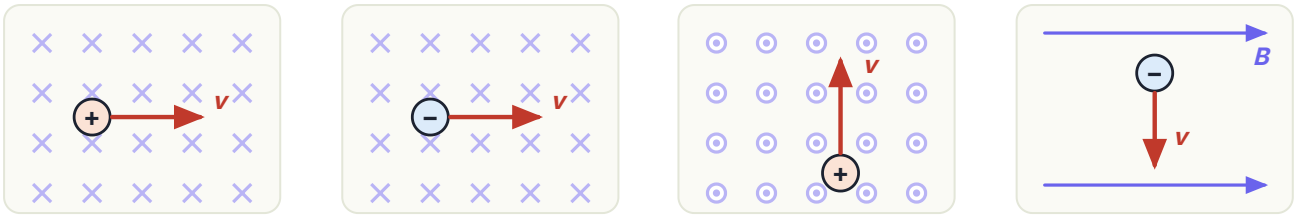
Figure 17.7: Fleming's left-hand rule. For a positive charge the second finger points along v; for a negative charge (electron) it points opposite to v.

Teacher's note

Some notes use the **right** hand for a negative charge. It gives the same answer, but it is easy to mix up with Fleming's right-hand rule for induction in Chapter 18. Safer: always use the left hand, with the second finger along the **conventional current** (opposite to the velocity of an electron).

Example 17.2.1

Find the direction of the magnetic force on the charge in each case.



(a) (b) (c) (d) electron

Solution

(a) $+q$ moving right, B into the page: **upwards** (towards the top of the page).

(b) $-q$ moving right, B into the page: the force is opposite to (a), **downwards**.

(c) $+q$ moving up, B out of the page: **to the right**.

(d) electron moving down, B to the right: for a positive charge it would be out of the page, so for the electron **into the page**.

Example 17.2.2

A proton travels at 3.1×10^7 m s^{-1} in a uniform field of 1.6 T. Find the force on it if its velocity (a) is perpendicular to the field, (b) makes 60° with the field. (e = 1.60×10^{-19} C)

Solution

(a) $F = qvB = (1.60 \times 10^{-19})(3.1 \times 10^7)(1.6) =$ **7.9×10^{-12} N**

(b) $F = qvB \sin 60° = (7.94 \times 10^{-12})(0.866) =$ **6.9×10^{-12} N**

Example 17.2.3

A charge q_1 = 25.0 μC moves at 4.5 × 10^3 m s^{-1} perpendicular to a uniform field and feels a force of 7.31 × 10^{-3} N. A second charge q_2 = 5.00 μC moves at 40.0° to the same field and feels 1.90 × 10^{-3} N. Find (a) B, (b) the speed of q_2.

Solution

(a) $B = \dfrac{F_1}{q_1 v_1} = \dfrac{7.31 \times 10^{-3}}{(25.0 \times 10^{-6})(4.5 \times 10^3)} = \mathbf{6.50 \times 10^{-2}\ T}$

(b) $v_2 = \dfrac{F_2}{q_2 B \sin 40.0°} = \dfrac{1.90 \times 10^{-3}}{(5.00 \times 10^{-6})(6.50 \times 10^{-2})(0.643)} = \mathbf{9.10 \times 10^3\ m\,s^{-1}}$

Example 17.2.4

An electron moves from left to right at 2.0 × 10^6 m s^{-1} into a field of 3.0 T directed vertically downwards (in the plane of the page). Find the magnitude and direction of the force on it.

Solution

$F = evB = (1.60 \times 10^{-19})(2.0 \times 10^6)(3.0) = \mathbf{9.6 \times 10^{-13}\ N}$

For a positive charge the force would be into the page, so on the electron it is **out of the page**.

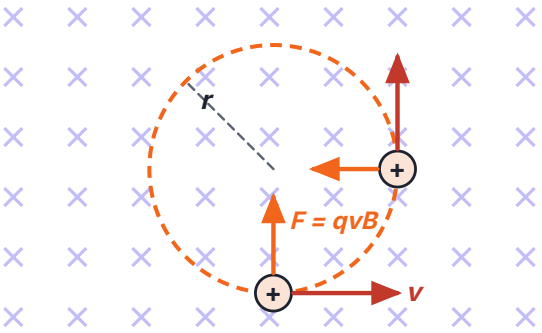
Motion parallel to the field

If **v** is parallel (or antiparallel) to **B**, sin θ = 0, so **there is no magnetic force**. The charge carries on in a **straight line at constant speed**.

Motion perpendicular to the field

The force is always perpendicular to the velocity, so it does **no work**: the speed and kinetic energy stay constant, only the direction changes.

A force of constant size always perpendicular to the velocity gives **uniform circular motion**. The magnetic force provides the centripetal force:



B into the page; a positive charge goes round anticlockwise

Figure 17.8: circular motion of a charge in a uniform magnetic field.

$$qvB = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB} \qquad T = \frac{2\pi r}{v} = \frac{2\pi m}{qB} \qquad f = \frac{qB}{2\pi m}$$

- A faster particle moves in a **larger** circle, but the **period does not depend on the speed**.
- A positive and a negative charge with the same speed curve in opposite directions.

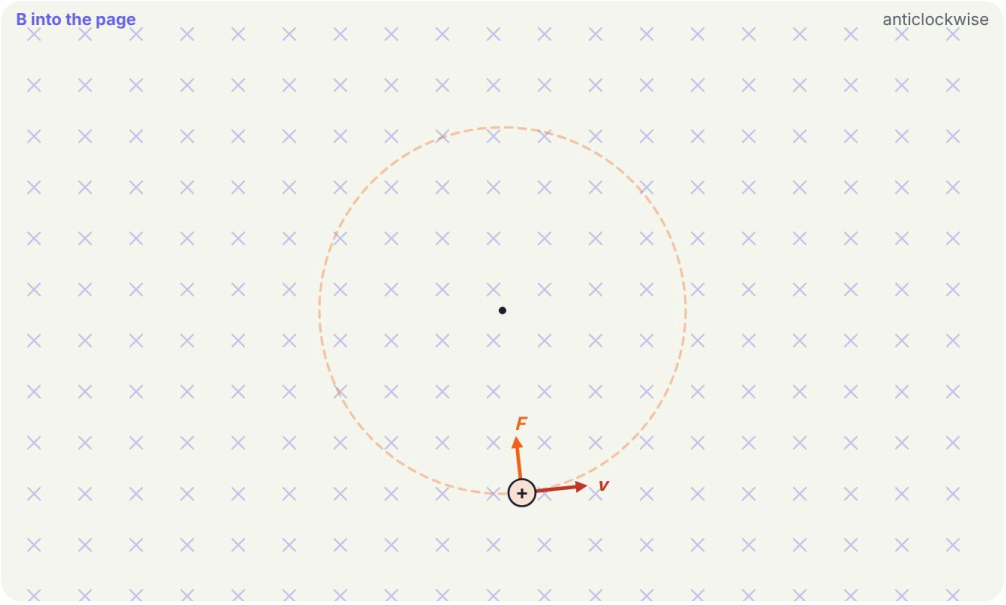
Teacher's note

If v makes any other angle with B, the component of v along B is unchanged and the perpendicular component goes round in a circle, so the path is a helix (a spiral along B).

Simulation

A charge in a uniform magnetic field

Choose the sign of the charge, its speed and the flux density. Perpendicular to B it moves in a circle: watch the radius change with v and B while the time for one turn depends only on B. Then switch to "parallel to B".



radius r = mv/(qB)	period T = 2πm/(qB)	magnetic force qvB	speed
13.9 cm	**1.093 µs**	**7.68×10^{-15} N**	**constant (F ⊥ v)**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.2.5

A proton moves at 3.0×10^5 m s^{-1} perpendicular to a field of 0.020 T. Find (a) its kinetic energy, (b) the magnetic force on it, (c) the radius of its path, (d) the period. ($m_p = 1.67 \times 10^{-27}$ kg)

Solution

(a) $K = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(1.67 \times 10^{-27})(3.0 \times 10^5)^2 = \mathbf{7.5 \times 10^{-17}\ J}$

(b) $F = qvB = (1.60 \times 10^{-19})(3.0 \times 10^5)(0.020) = \mathbf{9.6 \times 10^{-16}\ N}$

(c) $$r = \frac{mv}{qB} = \frac{(1.67 \times 10^{-27})(3.0 \times 10^5)}{(1.60 \times 10^{-19})(0.020)} = \mathbf{0.16\ m}$$

(d) $$T = \frac{2\pi m}{qB} = \frac{2\pi(1.67 \times 10^{-27})}{(1.60 \times 10^{-19})(0.020)} = \mathbf{3.3 \times 10^{-6}\ s}$$

Exercise: force on a moving charge

1. An electron enters a uniform field of 0.50 mT at 4.0×10^6 m s^{-1}, perpendicular to the field. Find the radius of its path. ($m_e = 9.11 \times 10^{-31}$ kg)

   ### Answer

   $r = mv/(eB)$ = **4.6 cm**

   FULL SOLUTION

   The magnetic force provides the centripetal force: $evB = mv^2/r$

   $r = mv/(eB) = (9.11 \times 10^{-31})(4.0 \times 10^6) / [(1.60 \times 10^{-19})(0.50 \times 10^{-3})]$

   $= 4.6 \times 10^{-2}$ m = **4.6 cm**

2. Explain why a magnetic field cannot change the kinetic energy of a moving charge.

   ### Answer

   The force is always perpendicular to the velocity, so it does no work. It changes the direction of motion but not the speed.

   FULL SOLUTION

   The magnetic force $F = qvB \sin\theta$ is always **perpendicular to the velocity**.

   So it has no component along the direction of motion and does **no work** ($W = Fs\cos 90° = 0$).

   By the work–energy theorem the kinetic energy does not change: the force changes the **direction** of motion but **not the speed**.

3. A proton and an alpha particle (charge $2e$, mass $4m$) enter the same field perpendicular to it with the same speed. Find the ratio of the radii $r_\alpha : r_p$.

   ### Answer

   $r \propto m/q$: **(4/2) : 1 = 2 : 1**

   FULL SOLUTION

   $qvB = mv^2/r \Rightarrow r = mv/(qB)$, so with the same v and B, $r \propto m/q$

   Proton: $m/q = m/e$; alpha particle: $m/q = 4m/2e = 2(m/e)$

   $r_\alpha : r_p = 2 : 1$

   = **2 : 1**

17.3 Force on a current-carrying conductor

A current is a stream of moving charges. Each charge feels a force qvB sin θ, and the forces on all the charges in the wire add up to a force on the wire.

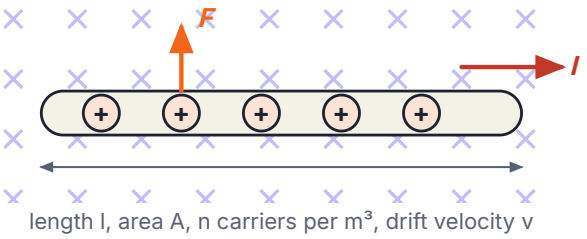


Figure 17.9: a length l of wire in a uniform field.

Deriving F = IlB sin θ

Number of carriers in the length l: $N = nAl$.

Force on each carrier: qvB sin θ, so the total force is $F = (nAl)qvB$ sin θ.

The current is $I = nAvq$ (Chapter 15), so $F = (nAvq)lB$ sin θ:

$$F = IlB \sin\theta \qquad \mathbf{F} = \boldsymbol{l} \times \mathbf{B}$$

θ = angle between the wire (current) and B. F = 0 when the wire is parallel to B; F = IlB when perpendicular.

- Direction: **Fleming's left-hand rule** with the second finger along the current.
- This is the principle of the electric motor and the moving-coil meter.

Example 17.3.1

A vertical wire is in a uniform field directed into the page. Find the direction of the force on it when the current flows (a) upwards, (b) downwards.



Solution

(a) **to the left** (b) **to the right** (reversing the current reverses the force)

Example 17.3.2

A wire 0.655 m long carries 21.0 A in a 0.470 T field and feels a force of 5.46 N. Find the angle (less than 90°) between the wire and the field.

Solution

$$\sin\theta = \frac{F}{IlB} = \frac{5.46}{(21.0)(0.655)(0.470)} = 0.845$$

θ = **57.6°**

Example 17.3.3

A square loop of side 0.32 m carries 12 A. It lies in the plane of a uniform 0.25 T field, with two sides perpendicular to the field and two sides parallel to it. Find the force on each side.

Solution

Sides perpendicular to B: $F = IlB = (12)(0.32)(0.25)$ = **0.96 N** each (in opposite directions)

Sides parallel to B: sin 0° = 0, so **no force**.

17.4 Magnetic fields due to currents

Ampère's law

the **line integral of B around any closed loop is equal to μ_0 times the current enclosed by the loop**.

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

μ_0 = permeability of free space = $4\pi \times 10^{-7}$ H m^{-1} (T m A^{-1})

Applying Ampère's law to a long straight wire

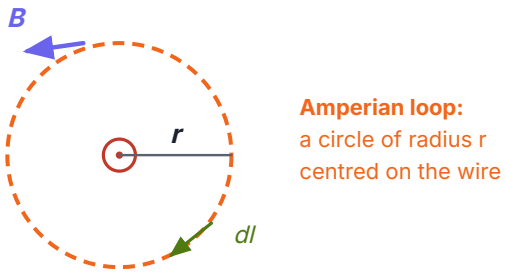


Figure 17.10: a circular Amperian loop around a long straight wire.

Choose a circle of radius r centred on the wire, in a plane perpendicular to it.

By symmetry B has the same size at every point on the circle and is tangential, so it is parallel to d**l** everywhere: **B**·d**l** = $B\,dl$.

$\oint B\,dl = B(2\pi r) = \mu_0 I$

$$B = \frac{\mu_0 I}{2\pi r}$$

long straight wire; B ∝ 1/r

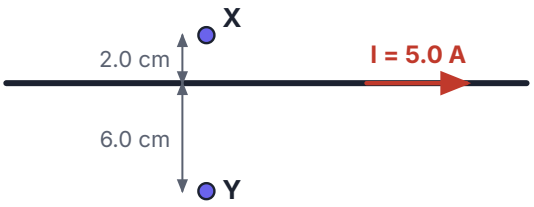
Circular coil and solenoid

Conductor	Flux density	Notes
Long straight wire	$B = \dfrac{\mu_0 I}{2\pi r}$	r = perpendicular distance from the wire
Flat circular coil, at its centre	$B = \dfrac{\mu_0 N I}{2r}$	N turns of radius r
Long solenoid, inside	$B = \mu_0 n I$	$n = \dfrac{N}{L}$ = turns per unit length

With a core of relative permeability μ_r, replace μ_0 by $\mu = \mu_r \mu_0$. A soft iron core makes the field of a solenoid much stronger.

Example 17.4.1

A long straight wire carries 5.0 A to the right. Point X is 2.0 cm above the wire and point Y is 6.0 cm below it. Find the flux density at X and at Y.



Solution

$$B_X = \frac{(4\pi \times 10^{-7})(5.0)}{2\pi(0.020)} = \mathbf{5.0 \times 10^{-5}\ T,\ out\ of\ the\ page}$$

$$B_Y = \frac{(4\pi \times 10^{-7})(5.0)}{2\pi(0.060)} = \mathbf{1.7 \times 10^{-5}\ T,\ into\ the\ page}\ \text{(right-hand grip rule)}$$

Example 17.4.2

A flat circular coil of 40 turns and radius 6.0 cm is in air. What current gives a flux density of 2.0 mT at its centre?

Solution

$$I = \frac{2rB}{\mu_0 N} = \frac{2(0.060)(2.0 \times 10^{-3})}{(4\pi \times 10^{-7})(40)} = \mathbf{4.8\ A}$$

Example 17.4.3

An air-cored solenoid has 2000 turns, length 60 cm and diameter 2.0 cm. Find the flux density inside it when the current is 5.0 A.

Solution

$$B = \mu_0 \frac{N}{L} I = (4\pi \times 10^{-7}) \frac{2000}{0.60} (5.0) = \mathbf{0.021\ T}$$ (the diameter is not needed)

Example 17.4.4

A solenoid of 400 turns is wound on an iron core 20 cm long with relative permeability 13 000. What current gives 0.50 T at its centre?

Solution

$$\mu = \mu_r \mu_0 = (13\,000)(4\pi \times 10^{-7}) = 1.63 \times 10^{-2}\ \text{H m}^{-1}$$

$$I = \frac{BL}{\mu N} = \frac{(0.50)(0.20)}{(1.63 \times 10^{-2})(400)} = \mathbf{0.015\ A}$$

Exercise: fields due to currents

1. Two long straight parallel wires are 30 cm apart and each carries 20 A. Find the flux density at a point between them, in their plane, 10 cm from one wire and 20 cm from the other, when the currents are (a) in the same direction, (b) in opposite directions.

Answer

The fields of the wires are 4.0×10^{-5} T and 2.0×10^{-5} T. (a) They are opposite: $\mathbf{2.0 \times 10^{-5}}$ T, in the direction of the nearer wire's field. (b) They add: $\mathbf{6.0 \times 10^{-5}}$ T. Both perpendicular to the plane of the wires.

FULL SOLUTION

$B = \mu_0 I/(2\pi r)$ for a long straight wire.

Nearer wire ($r = 0.10$ m): $B_1 = (4\pi \times 10^{-7})(20) / (2\pi \times 0.10) = 4.0 \times 10^{-5}$ T

Farther wire ($r = 0.20$ m): $B_2 = (4\pi \times 10^{-7})(20) / (2\pi \times 0.20) = 2.0 \times 10^{-5}$ T

(a) Same direction: between the wires the two fields are in opposite directions.

$B = 4.0 \times 10^{-5} - 2.0 \times 10^{-5} = \mathbf{2.0 \times 10^{-5}}$ T, in the direction of the nearer wire's field

(b) Opposite directions: between the wires the two fields are in the same direction.

$B = 4.0 \times 10^{-5} + 2.0 \times 10^{-5} = \mathbf{6.0 \times 10^{-5}}$ T

In both cases the field is **perpendicular to the plane of the wires**.

2. How far from a long straight wire carrying 12 A is the flux density 3.0×10^{-5} T?

Answer

$r = \mu_0 I/(2\pi B)$ = **8.0 cm**

FULL SOLUTION

$B = \mu_0 I/(2\pi r) \Rightarrow r = \mu_0 I/(2\pi B)$

$r = (4\pi \times 10^{-7})(12) / (2\pi \times 3.0 \times 10^{-5})$

= 0.080 m = **8.0 cm**

17.5 Force between two current-carrying conductors

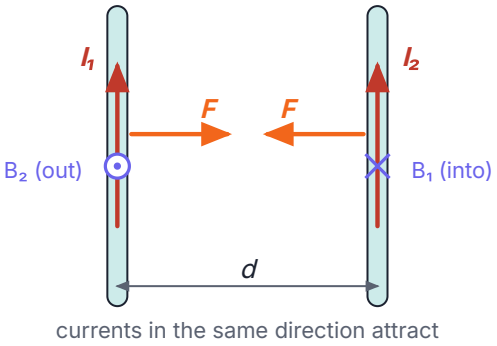


Figure 17.11: two long parallel wires a distance d apart.

Derivation

Wire 1 produces a field at wire 2: $B_1 = \dfrac{\mu_0 I_1}{2\pi d}$, perpendicular to wire 2.

A length l of wire 2 feels $F = B_1 I_2 l \sin 90°$.

So $F = \dfrac{\mu_0 I_1 I_2 l}{2\pi d}$. Working the other way (field of wire 2 at wire 1) gives a force of the same size on wire 1, in the opposite direction (Newton's third law).

$$F = \frac{\mu_0 I_1 I_2 l}{2\pi d} \qquad \frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

currents in the **same direction attract**; in **opposite directions repel**

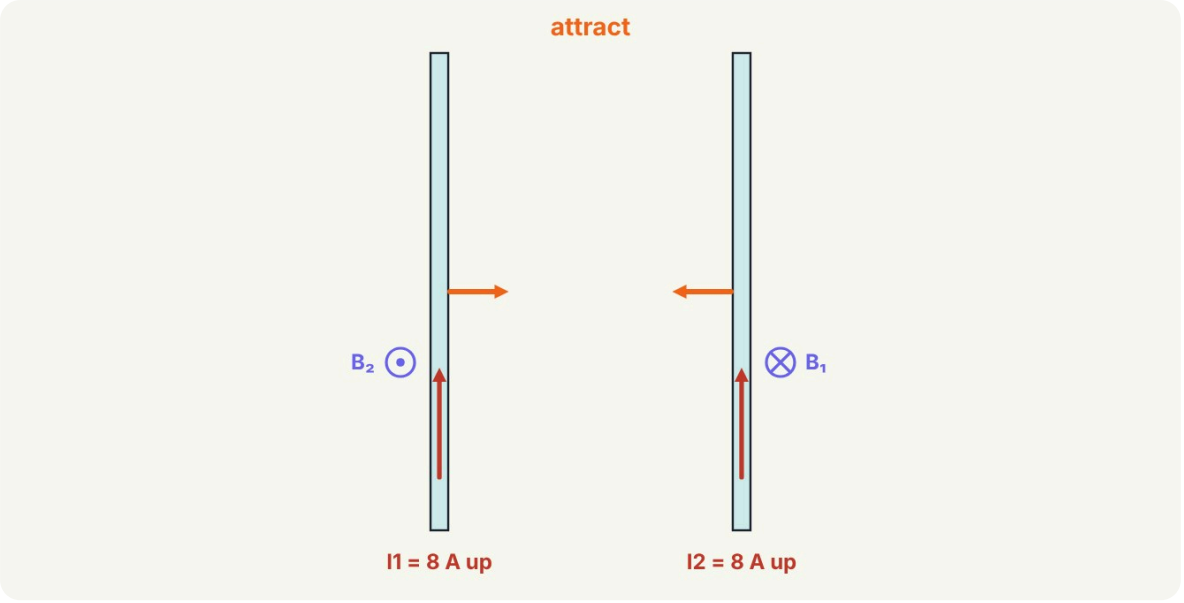
- With $I_1 = I_2 = 1$ A and $d = 1$ m: $F/l = 2 \times 10^{-7}$ N m^{-1}. This was used to define the ampere: the constant current which, in each of two infinitely long, straight, parallel conductors of negligible cross-section 1 m apart in a vacuum, gives a force of 2×10^{-7} N per metre of length.

Teacher's note

Since 2019 the SI defines the ampere by fixing the value of the elementary charge, but this result is still true and is a favourite exam question.

Simulation Two parallel currents

Change the size and direction of each current and the separation. See the field of each wire at the other one and the forces: same direction attract, opposite directions repel. The forces are always equal and opposite.



B_1 at wire 2 = $\mu_0 I_1/(2\pi d)$	force per metre $F/l = \mu_0 I_1 I_2/(2\pi d)$	B_2 at wire 1	forces on the wires
4.00×10^{-5} T	**3.20×10^{-4} N m^{-1}**	**4.00×10^{-5} T**	**equal and opposite**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.5.1

Two long parallel wires 2.0 cm apart in air carry 8.0 A and 10 A. Find the force on each metre of wire.

Solution

$$\frac{F}{l} = \frac{(4\pi \times 10^{-7})(8.0)(10)}{2\pi(0.020)} = \mathbf{8.0 \times 10^{-4}\ N\ m^{-1}}$$

Example 17.5.2

Two long parallel wires 5.0 cm apart attract each other with a force of 6.0×10^{-7} N per metre. One carries 400 mA. Find (a) the current in the other, (b) its direction relative to the first.

Solution

(a) $I_2 = \dfrac{(F/l)(2\pi d)}{\mu_0 I_1} = \dfrac{(6.0 \times 10^{-7})(2\pi)(0.050)}{(4\pi \times 10^{-7})(0.400)} = \mathbf{0.375\ A}$

(b) They attract, so the currents are in the **same direction**.

17.6 Determination of the ratio e/m

Crossed electric and magnetic fields

When **E**, **B** and **v** are perpendicular to each other, the electric and magnetic forces on a charge are in opposite directions. If they are equal, the resultant force is zero and the charge goes **straight through, undeflected**.

$$qE = qvB \Rightarrow v = \frac{E}{B}$$

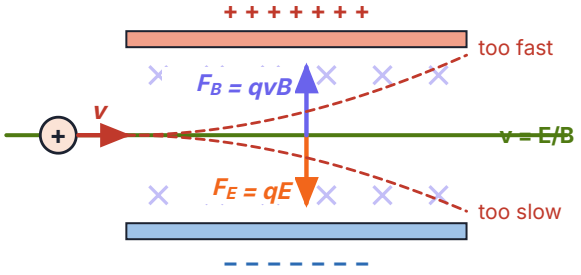


Figure 17.12: a velocity selector. Only particles with v = E/B pass straight through; the result does not depend on q or m.

- Faster particles: $qvB > qE$, they bend towards the magnetic force. Slower particles bend towards the electric force.
- This arrangement is a **velocity selector** (used, for example, before the magnetic field of a mass spectrometer).

Thomson's experiment

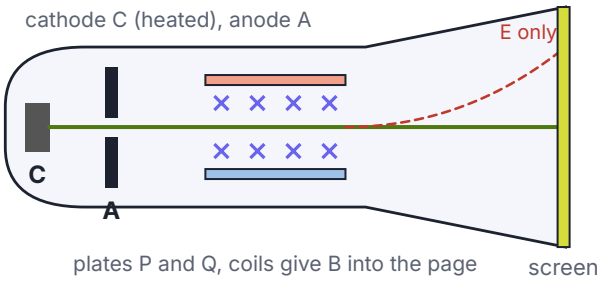


Figure 17.13: Thomson's apparatus (simplified).

Electrons from a heated cathode C are accelerated to the anode A and pass through it as a narrow beam.

Between plates P and Q there is an electric field E; coils outside the tube give a magnetic field B at right angles to it and to the beam.

E (or B) is adjusted until the beam is **undeflected**: then $eE = evB$, so the speed is $v = \dfrac{E}{B}$.

The electric field is switched off. In the magnetic field alone the beam moves in a circular arc of radius r, found from the deflection on the screen.

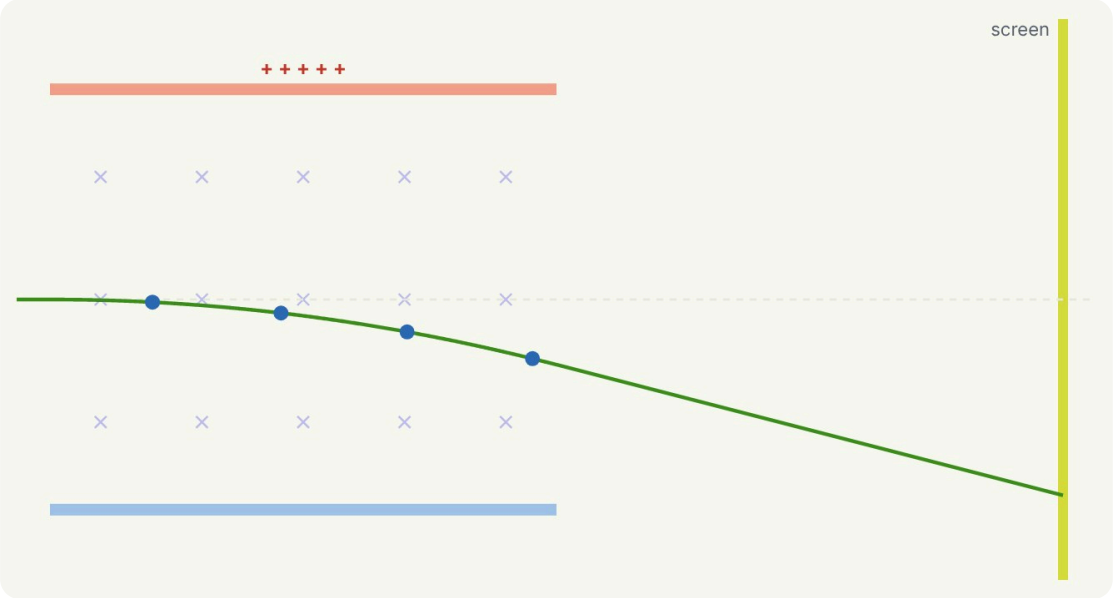
The magnetic force provides the centripetal force: $evB = \dfrac{mv^2}{r}$, so $\dfrac{e}{m} = \dfrac{v}{Br}$. Substituting $v = E/B$:

$$\frac{e}{m} = \frac{E}{B^2 r}$$

Thomson found the same e/m for any cathode material: electrons are part of every atom. Accepted value 1.76×10^{11} C kg^{-1}.

Simulation Crossed fields and Thomson's e/m

Fire electrons through the plates. Adjust E and B until the beam goes straight through (v = E/B). Then switch off E: the beam bends into a circle, and its radius gives e/m.



v needed = E/B
1.40 × 10⁷ m s⁻¹

beam
bends down: evB > eE (too fast)

electric force eE
2.24 × 10⁻¹⁵ N

magnetic force evB
3.20 × 10⁻¹⁵ N

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.6.1

Protons pass undeflected through crossed fields with E = 3.0 × 10^5 V m^{-1} and B = 0.25 T. Find their speed.

Solution

$$v = \frac{E}{B} = \frac{3.0 \times 10^5}{0.25} = \mathbf{1.2 \times 10^6\ m\,s^{-1}}$$

Example 17.6.2

A velocity selector must let positive ions moving to the right at 6.0 km s^{-1} through undeflected. The electric field is 300 V m^{-1} into the page. Find the size and direction of the magnetic field.

Solution

$$B = \frac{E}{v} = \frac{300}{6.0 \times 10^3} = \mathbf{0.050\ T}$$

The electric force on the ions is into the page, so the magnetic force must be out of the page. By Fleming's left-hand rule (second finger to the right, thumb out of the page) B points **up the page**.

Example 17.6.3

In a Thomson experiment the beam is undeflected with $E = 2.0 \times 10^4$ V m^{-1} and $B = 1.0$ mT. With the electric field switched off, the beam moves in an arc of radius 11.4 cm. Find (a) the speed of the electrons, (b) e/m.

Solution

(a) $$v = \frac{E}{B} = \frac{2.0 \times 10^4}{1.0 \times 10^{-3}} = \mathbf{2.0 \times 10^7\ m\,s^{-1}}$$

(b) $$\frac{e}{m} = \frac{E}{B^2 r} = \frac{2.0 \times 10^4}{(1.0 \times 10^{-3})^2 (0.114)} = \mathbf{1.75 \times 10^{11}\ C\ kg^{-1}}$$

A real use: the mass spectrometer

Atoms of the sample are **ionised** (given the same charge, usually $+e$) and accelerated into a beam.

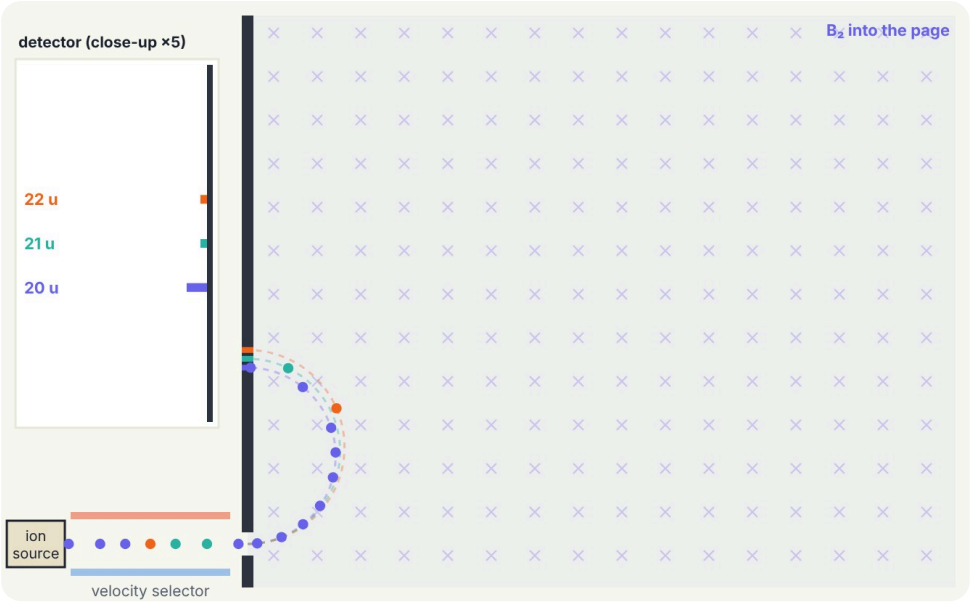
A **velocity selector** (crossed E and B_1) lets through only ions with $v = E/B_1$, so all ions leave it with the same speed.

In a second field B_2 alone, each ion moves in a semicircle of radius $r = mv/(qB_2) \propto m$: heavier ions make bigger circles and land further along the detector.

So the detector separates **isotopes** (same charge, different mass). It is used to date rocks and fossils (carbon-14), test for drugs in sport and check the purity of chemicals.

Simulation Mass spectrometer

Choose an element and watch its isotopes leave the velocity selector at the same speed and then separate in the magnetic field B_2. Change B_2: a stronger field makes smaller circles and brings the isotopes closer together on the detector.



speed from the selector $v = E/B_1$	r for mass 20 u	r for mass 21 u	r for mass 22 u
1.0×10^5 m s^{-1}	**4.15 cm**	**4.36 cm**	**4.56 cm**

gap between the outer two on the detector
8.3 mm

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.6.4

Singly charged neon ions leave a velocity selector at 1.0×10^5 m s^{-1} and enter a field $B_2 = 0.50$ T. Find the distance between the places where neon-20 and neon-22 ions hit the detector. (1 u = 1.66×10^{-27} kg)

Solution

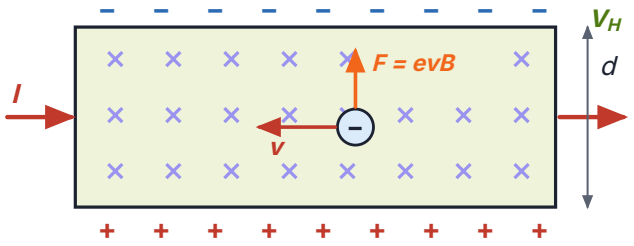
$$r_{20} = \frac{mv}{eB} = \frac{(20 \times 1.66 \times 10^{-27})(1.0 \times 10^5)}{(1.60 \times 10^{-19})(0.50)} = 4.15 \times 10^{-2} \text{ m; similarly } r_{22} = 4.57 \times 10^{-2} \text{ m}$$

Each ion lands a diameter ($2r$) from the entrance slit, so the separation is $2(r_{22} - r_{20}) = 2(0.42 \times 10^{-2}) =$ **8.3×10^{-3} m** (8.3 mm)

17.7 Hall effect

Hall effect

when a current flows through a conductor or semiconductor in a magnetic field perpendicular to the current, **a p.d. (the Hall voltage) appears across the sides of the specimen**, perpendicular to both the current and the field.



top view: B into the page; t = thickness along B

Figure 17.14: the Hall effect in a slab with electrons as charge carriers.

How the Hall voltage builds up

The charge carriers drift along the slab with velocity v. The magnetic force evB pushes them towards one side.

Charges collect on that side and an equal opposite charge is left on the other side. This sets up an electric field E_H across the slab.

The electric force eE_H opposes the magnetic force. Charges keep collecting until the two forces balance; after that the carriers flow straight along the slab and the Hall voltage is steady.

Deriving the Hall voltage

At equilibrium: $eE_H = evB$, and $E_H = \frac{V_H}{d}$ (d = width across which V_H appears), so $V_H = Bvd$.

The current is $I = nevA$ with $A = td$ (t = thickness, measured along B), so $v = \frac{I}{netd}$.

$V_H = B\frac{I}{netd}d$:

$$V_H = \frac{BI}{net}$$

n = number of charge carriers per unit volume, t = thickness of the specimen in the direction of B

- $V_H \propto 1/n$. Metals have a very large n, so their Hall voltage is tiny. Semiconductors have a much smaller n, so V_H is large enough to measure: Hall probes use semiconductors.
- A thin specimen (small t) gives a larger V_H.

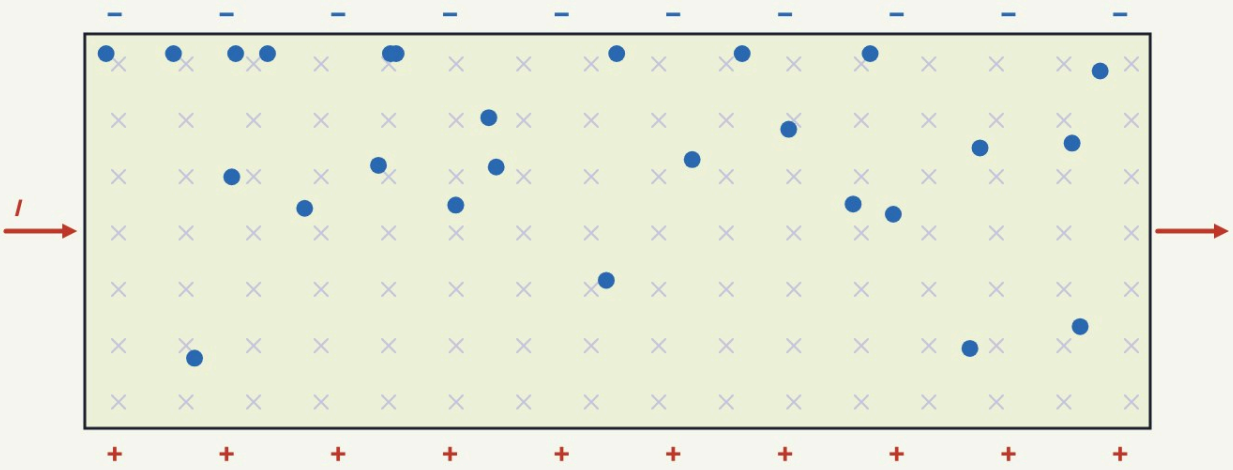
Applications of the Hall effect

- **Measuring magnetic flux density** with a Hall probe: with I constant, $V_H \propto B$. The probe is small, so it can measure B at a point, and it responds quickly to changing fields.
- **Finding the sign of the charge carriers**: the side that becomes positive shows whether the carriers are electrons (n-type) or holes (p-type).
- **Finding the number density** n of the carriers: $n = BI/(eV_H t)$.
- Hall sensors in everyday devices: measuring current without breaking the circuit, detecting the speed of a rotating wheel, and as contactless switches.

Simulation

Hall effect

Choose electrons or holes as the charge carriers and change B and the current. Watch the carriers pile up on one face until the Hall voltage stops further build-up. The sign of the top face tells you the sign of the carriers.



top face: negative

V_H = 3.12 mV

Hall voltage V_H = BIv(net)	carriers pushed to	n (this slab)	thickness t
3.12 mV	**the top face**	**1.0×10^{22} m^{-3}**	**1.0 mm**

This is a still picture. Open PhysiQ › Notes › Semester 2 › Chapter 17 to use the simulation.

Example 17.7.1

A thin semiconductor slab is 10.0 mm long, 3.0 mm wide and 1.0 mm thick, with 4.0×10^{21} charge carriers per m^3. A current of 0.0025 mA flows along its length. A uniform field B is applied perpendicular to its largest face (along the 1.0 mm thickness), and a Hall voltage of 60 μV appears across its width. Find (a) the drift velocity of the carriers, (b) B. (c) The top face (as in Figure 17.14) becomes negative. What are the charge carriers?

Solution

(a) $v = \dfrac{I}{neA} = \dfrac{2.5 \times 10^{-6}}{(4.0 \times 10^{21})(1.60 \times 10^{-19})(3.0 \times 10^{-3} \times 1.0 \times 10^{-3})} = \mathbf{1.3 \times 10^{-3}\ m\,s^{-1}}$

(b) $B = \dfrac{V_H net}{I} = \dfrac{(60 \times 10^{-6})(4.0 \times 10^{21})(1.60 \times 10^{-19})(1.0 \times 10^{-3})}{2.5 \times 10^{-6}} = \mathbf{15\ T}$

(c) With the current to the right and B into the page, positive carriers moving right and negative carriers moving left are both pushed to the top face. The top face is negative, so the carriers are **electrons** (n-type).

Exercise: crossed fields and the Hall effect

1. Why does a Hall probe use a semiconductor and not a metal?

   Answer

   $V_H = BI/(net)$ is inversely proportional to n. A metal has so many free electrons that V_H is too small to measure; a semiconductor has far fewer carriers, so V_H is much larger.

   FULL SOLUTION

   Hall voltage $V_H = BI/(nte)$, so $V_H \propto 1/n$.

   A metal has a very large number of free electrons per unit volume, so V_H is **too small to measure**.

   A semiconductor has far fewer charge carriers, so **V_H is much larger** and easy to measure.

2. Electrons accelerated to 3.0×10^7 m s^{-1} enter crossed fields with B = 2.0 mT. What electric field lets them pass undeflected?

   Answer

   $E = vB = 6.0 \times 10^4$ V m^{-1}

   FULL SOLUTION

   Undeflected: electric force = magnetic force, eE = evB

   $E = vB = (3.0 \times 10^7)(2.0 \times 10^{-3})$

$= 6.0 \times 10^4\ \text{V m}^{-1}$

3. Derive $e/m = E/(B^2 r)$ for Thomson's experiment.

Answer

Undeflected: $eE = evB$ so $v = E/B$. Magnetic field only: $evB = mv^2/r$, so $e/m = v/(Br) = E/(B^2 r)$.

FULL SOLUTION

With both fields adjusted so the beam is undeflected: eE = evB ⇒ v = E/B

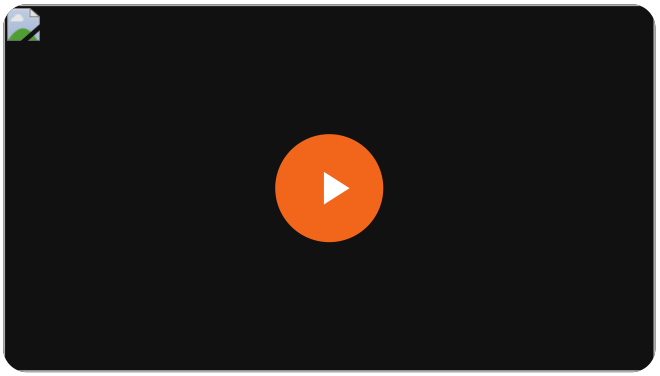
With the magnetic field only, the beam moves in a circle of radius r: evB = mv^2/r ⇒ e/m = v/(Br)

Substitute v = E/B:

e/m = E/(B^2r)

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



REAL DEMONSTRATION

Demonstrating the power of MRI magnets

Open in YouTube ↗

The magnet inside an MRI scanner

A current in a superconducting solenoid produces a very strong uniform magnetic field ($B = \mu_o nI$); steel objects are pulled in with great force.